

FLUIDS

1. Substances that can flow are called

- A) Solids
- B) Fluids
- C) Metals
- D) Gases only

Answer: B

2. Fluids include

- A) Solids
- B) Liquids and gases
- C) Metals
- D) Rocks

Answer: B

3. Pressure is defined as

- A) Force per unit area
- B) Force per volume
- C) Mass per area
- D) Energy per area

Answer: A

4. SI unit of pressure

- A) Pascal
- B) Joule
- C) Newton
- D) Watt

Answer: A

5. Pressure increases when

- A) Area increases
- B) Force increases
- C) Force decreases
- D) Area increases and force decreases

Answer: B

6. Pressure decreases when

- A) Area increases
- B) Area decreases

C) Force increases

D) Mass increases

Answer: A

7. Pressure formula is

A) $P = F/A$

B) $P = A/F$

C) $P = F \times A$

D) $P = F + A$

Answer: A

8. Atmospheric pressure is caused by

A) Weight of air

B) Weight of water

C) Weight of soil

D) Weight of metal

Answer: A

9. Instrument used to measure atmospheric pressure

A) Barometer

B) Thermometer

C) Hydrometer

D) Ammeter

Answer: A

10. Mercury barometer was invented by

A) Torricelli

B) Newton

C) Galileo

D) Einstein

Answer: A

11. Standard atmospheric pressure is

A) 760 mm Hg

B) 500 mm Hg

C) 100 mm Hg

D) 1000 mm Hg

Answer: A

12. Pressure in liquids increases with

- A) Depth
- B) Height
- C) Area
- D) Volume

Answer: A

13. Pressure at the same depth in a liquid is

- A) Equal
- B) Unequal
- C) Zero
- D) Infinite

Answer: A

14. The force exerted by fluid on surface is

- A) Pressure
- B) Density
- C) Mass
- D) Volume

Answer: A

15. Density is defined as

- A) Mass per unit volume
- B) Volume per mass
- C) Area per mass
- D) Mass per area

Answer: A

16. Density formula

- A) Density = Mass/Volume
- B) Density = Volume/Mass
- C) Density = Mass×Volume
- D) Density = Mass+Volume

Answer: A

17. SI unit of density

- A) kg/m^3
- B) kg
- C) m^3

D) N

Answer: A

18. Liquids exert pressure

A) In all directions

B) Upwards only

C) Downwards only

D) Sideways only

Answer: A

19. Principle explaining floating of bodies

A) Archimedes principle

B) Newton law

C) Boyle law

D) Hooke law

Answer: A

20. Archimedes principle states that

A) Upward thrust equals weight of displaced liquid

B) Objects sink always

C) Liquids do not exert pressure

D) Solids float always

Answer: A

21. Upward force exerted by fluid is called

A) Buoyant force

B) Gravitational force

C) Magnetic force

D) Electric force

Answer: A

22. Buoyant force acts

A) Upwards

B) Downwards

C) Sideways

D) Diagonally

Answer: A

23. Object floats when

A) Buoyant force $\geq$ weight

- B) Weight greater than buoyant force
- C) Weight equal to zero
- D) Density zero

Answer: A

24. Object sinks when

- A) Weight greater than buoyant force
- B) Buoyant force greater than weight
- C) Weight zero
- D) Density zero

Answer: A

25. Density of water

- A) 1000 kg/m^3
- B) 500 kg/m^3
- C) 2000 kg/m^3
- D) 100 kg/m^3

Answer: A

26. Relative density is ratio of

- A) Density of substance to density of water
- B) Density of water to density of substance
- C) Mass to volume
- D) Volume to mass

Answer: A

27. Relative density has

- A) No unit
- B) Pascal
- C) kg
- D) N

Answer: A

28. Hydrometer is used to measure

- A) Relative density of liquids
- B) Temperature
- C) Pressure
- D) Mass

Answer: A

29. Lactometer measures

- A) Purity of milk
- B) Pressure
- C) Temperature
- D) Volume

Answer: A

30. Buoyancy depends on

- A) Density of fluid
- B) Volume displaced
- C) Gravity
- D) All of these

Answer: D

31. Ships float because

- A) Average density less than water
- B) Density more than water
- C) Weight zero
- D) Volume zero

Answer: A

32. Submarines float or sink by

- A) Changing density
- B) Changing mass
- C) Changing volume only
- D) Changing colour

Answer: A

33. Hydraulic machines work based on

- A) Pascal law
- B) Newton law
- C) Ohm law
- D) Boyle law

Answer: A

34. Pascal law states

- A) Pressure applied to fluid transmitted equally
- B) Pressure disappears
- C) Pressure only downward

D) Pressure only upward

Answer: A

35. Example of Pascal law application

A) Hydraulic brake

B) Thermometer

C) Barometer

D) Compass

Answer: A

36. Hydraulic press is used for

A) Lifting heavy loads

B) Heating liquids

C) Measuring pressure

D) Measuring density

Answer: A

37. Pressure depends on

A) Force and area

B) Mass and volume

C) Density only

D) Temperature

Answer: A

38. Greater area gives

A) Less pressure

B) More pressure

C) Same pressure

D) Zero pressure

Answer: A

39. Sharp knife cuts easily because

A) Small area produces high pressure

B) Large area produces high pressure

C) No pressure

D) Zero force

Answer: A

40. Camel walks easily on sand because

A) Large feet area

- B) Small feet area
- C) Heavy body
- D) Long neck

Answer: A

41. Atmospheric pressure at sea level is about

- A) 1 atm
- B) 2 atm
- C) 5 atm
- D) 10 atm

Answer: A

42. Atmospheric pressure decreases with

- A) Height
- B) Depth
- C) Temperature
- D) Mass

Answer: A

43. Drinking straw works due to

- A) Atmospheric pressure
- B) Magnetic force
- C) Electric force
- D) Gravity only

Answer: A

44. Syringe works due to

- A) Pressure difference
- B) Heat
- C) Light
- D) Sound

Answer: A

45. Density of object determines

- A) Floating or sinking
- B) Colour
- C) Temperature
- D) Shape

Answer: A

46. Ice floats on water because

- A) Density less than water
- B) Density greater
- C) Mass zero
- D) Pressure zero

Answer: A

47. Pressure in gases acts

- A) In all directions
- B) Downward only
- C) Upward only
- D) Sideways only

Answer: A

48. Liquids have

- A) Definite volume
- B) Definite shape
- C) No volume
- D) No mass

Answer: A

49. Gases have

- A) No definite shape and volume
- B) Definite shape
- C) Definite volume
- D) Zero mass

Answer: A

50. Fluid pressure increases with

- A) Depth
- B) Height
- C) Area
- D) Volume

Answer: A

51. Unit of force

- A) Newton
- B) Pascal
- C) Joule

D) Watt

Answer: A

52. Pressure is measured using

A) Manometer

B) Thermometer

C) Voltmeter

D) Ammeter

Answer: A

53. Buoyant force depends on

A) Density of fluid

B) Volume displaced

C) Gravity

D) All of these

Answer: D

54. Floating objects displace

A) Liquid equal to weight

B) Air only

C) Gas only

D) Metal only

Answer: A

55. Archimedes discovered principle while

A) Bathing

B) Studying

C) Sleeping

D) Walking

Answer: A

56. SI unit of buoyant force

A) Newton

B) Pascal

C) Joule

D) Watt

Answer: A

57. Fluids transmit pressure

A) Equally

- B) Unequally
- C) Only upward
- D) Only downward

Answer: A

58. Hydraulic lift uses

- A) Pascal law
- B) Newton law
- C) Boyle law
- D) Hooke law

Answer: A

59. Fluid pressure acts

- A) Perpendicular to surface
- B) Parallel to surface
- C) Diagonal
- D) Horizontal only

Answer: A

60. Water pressure increases with

- A) Depth
- B) Height
- C) Width
- D) Length

Answer: A

61. Dam walls are thicker at

- A) Bottom
- B) Top
- C) Middle
- D) Same thickness

Answer: A

62. This is because

- A) Pressure greater at bottom
- B) Pressure zero at bottom
- C) Pressure less at bottom
- D) Pressure same everywhere

Answer: A

63. Density of air is

- A) Less than water
- B) Equal to water
- C) Greater than water
- D) Infinite

Answer: A

64. Oil floats on water because

- A) Density less than water
- B) Density greater
- C) Same density
- D) Zero density

Answer: A

65. Pressure due to liquids depends on

- A) Density
- B) Depth
- C) Gravity
- D) All of these

Answer: D

66. Fluid mechanics studies

- A) Fluids
- B) Solids
- C) Metals
- D) Rocks

Answer: A

67. Air exerts

- A) Pressure
- B) Heat
- C) Light
- D) Sound

Answer: A

68. Atmospheric pressure acts

- A) In all directions
- B) Upward only
- C) Downward only

D) Sideways only

Answer: A

69. Mountaineers carry oxygen because

A) Air pressure low

B) Air pressure high

C) Air density high

D) Air heavy

Answer: A

70. Boiling point decreases at high altitude because

A) Low pressure

B) High pressure

C) No pressure

D) Same pressure

Answer: A

71. Relative density of water

A) 1

B) 0

C) 10

D) 100

Answer: A

72. Objects with relative density less than 1

A) Float

B) Sink

C) Dissolve

D) Evaporate

Answer: A

73. Objects with relative density greater than 1

A) Sink

B) Float

C) Evaporate

D) Dissolve

Answer: A

74. Fluid includes

A) Liquids and gases

B) Solids

C) Metals

D) Rocks

Answer: A

75. Example of fluid

A) Water

B) Iron

C) Wood

D) Stone

Answer: A

76. Gas molecules move

A) Freely

B) Fixed

C) Vibrate only

D) Do not move

Answer: A

77. Liquids take shape of

A) Container

B) Fixed shape

C) Sphere

D) Cube

Answer: A

78. Pressure unit Pascal equals

A) N/m^2

B) N/m

C) kg/m

D) kg/m^2

Answer: A

79. Pressure acts

A) Perpendicular

B) Parallel

C) Horizontal

D) Diagonal

Answer: A

80. Buoyancy helps

- A) Ships float
- B) Objects sink
- C) Air disappear
- D) Water vanish

Answer: A

81. Fluid pressure is measured by

- A) Manometer
- B) Thermometer
- C) Barometer
- D) Voltmeter

Answer: A

82. Hydraulics deals with

- A) Fluid pressure
- B) Heat
- C) Electricity
- D) Light

Answer: A

83. Pressure increases when area

- A) Decreases
- B) Increases
- C) Same
- D) Zero

Answer: A

84. Pressure decreases when force

- A) Decreases
- B) Increases
- C) Same
- D) Doubles

Answer: A

85. SI unit of volume

- A) m^3
- B) kg
- C) N

D) Pa

Answer: A

86. Density of steel is

A) Greater than water

B) Less than water

C) Equal to water

D) Zero

Answer: A

87. Steel ships float because

A) Overall density less than water

B) Density greater

C) Weight zero

D) Pressure zero

Answer: A

88. Pressure in liquids is due to

A) Weight of liquid

B) Air

C) Heat

D) Light

Answer: A

89. Gas pressure is due to

A) Collision of molecules

B) Gravity

C) Heat

D) Light

Answer: A

90. Atmospheric pressure supports

A) Mercury column

B) Water column

C) Air column

D) Metal column

Answer: A

91. Torricelli experiment proved

A) Atmospheric pressure exists

- B) Air has no weight
- C) Liquids have no pressure
- D) Water has no density

Answer: A

92. Pressure inside fluid increases with

- A) Depth
- B) Height
- C) Area
- D) Volume

Answer: A

93. Fluid pressure acts equally in

- A) All directions
- B) Upward
- C) Downward
- D) Sideways

Answer: A

94. Submarine works on

- A) Buoyancy principle
- B) Gravity principle
- C) Electric principle
- D) Magnetic principle

Answer: A

95. Balloon rises because

- A) Density less than air
- B) Density greater
- C) Mass zero
- D) Pressure zero

Answer: A

96. Fluid mechanics is part of

- A) Physics
- B) Chemistry
- C) Biology
- D) Geography

Answer: A

97. Liquids are

- A) Nearly incompressible
- B) Highly compressible
- C) Empty
- D) Solid

Answer: A

98. Gases are

- A) Highly compressible
- B) Incompressible
- C) Solid
- D) Fixed

Answer: A

99. Fluids flow due to

- A) Weak intermolecular forces
- B) Strong bonds
- C) Fixed particles
- D) No particles

Answer: A

100. Study of fluids helps in

- A) Engineering
- B) Medicine
- C) Aviation
- D) All of these

Answer: D